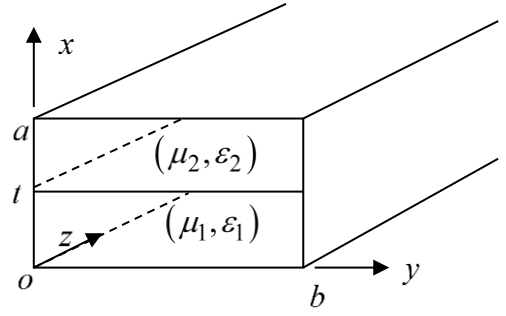


Class Notes 12

- A partially dielectric-filled rectangular waveguide

We cannot solve this problem TM_z or TE_z .

To see why, attempt to solve TM_z , set $\mathbf{A} = \psi \hat{\mathbf{z}}$



$$\mathbf{E} = \frac{-j\omega}{k^2} \left(k^2 \psi \hat{\mathbf{z}} + \nabla \frac{\partial \psi}{\partial z} \right)$$

$$E_x = \frac{-j\omega}{k^2} \frac{\partial^2 \psi}{\partial x \partial z} \quad (1)$$

$$E_y = \frac{-j\omega}{k^2} \frac{\partial \psi}{\partial y \partial z} \quad (2)$$

$$E_z = \frac{-j\omega}{k^2} \left(k^2 + \frac{\partial^2}{\partial z^2} \right) \psi \quad (3)$$

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \psi \hat{\mathbf{z}}$$

$$H_x = \frac{1}{\mu} \frac{\partial \psi}{\partial y} \quad (4)$$

$$H_y = -\frac{1}{\mu} \frac{\partial \psi}{\partial x} \quad (5)$$

$$H_z = 0 \quad (6)$$

In the region $x \in (0, t)$

$$\psi_1 = X_1 Y_1 Z_1$$

$$(\nabla^2 + k_1^2) \psi_1 = 0$$

$$k_{x1}^2 + k_{y1}^2 + k_{z1}^2 = k_1^2 = \omega^2 \mu_1 \epsilon_1$$

In the region $x \in (t, a)$

$$\psi_2 = X_2 Y_2 Z_2$$

$$(\nabla^2 + k_2^2) \psi_2 = 0$$

$$k_{x2}^2 + k_{y1}^2 + k_{z1}^2 = k_2^2 = \omega^2 \mu_2 \epsilon_2$$

Note that according to Snell's law $k_{y1} = k_{y2}$ and $k_{z1} = k_{z2}$, so let

$$k_{y1} = k_{y2} \doteq k_y$$

$$k_{z1} = k_{z2} \doteq k_z$$

Also since $Y_{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}$ must satisfy the same B.C.'s at $y=0$ and $y=b$ (i.e. $Y=0$), then

$$Y_{\begin{pmatrix} 1 \\ 2 \end{pmatrix}} = \sin\left(\frac{n\pi}{b} y\right).$$

And, because there are no z-direction B.C.'s,

$$Z_{\begin{pmatrix} 1 \\ 2 \end{pmatrix}} = e^{\mp jk_z z}.$$

However, the X 's must satisfy different B.C.'s and therefore must have different solutions in the two regions.

Choose:

$$X_1 = \sin(k_{x1}x) \text{ which insures that } \psi_1 = 0 \text{ at } x = 0 \text{ in region 1}$$

$$X_2 = \sin[k_{x2}(a-x)] \text{ which insures that } \psi_2 = 0 \text{ at } x = a \text{ in region 2}$$

Therefore $\psi = XYZ$ becomes:

In the region $x \in (0, t)$

In the region $x \in (t, a)$

$$\psi_1 = C_n^{1\pm} \sin(k_{x1}x) \sin\left(\frac{n\pi}{b}y\right) e^{\mp jk_z z}$$

$$\psi_2 = C_n^{2\pm} \sin k_{x2}(a-x) \sin\left(\frac{n\pi}{b}y\right) e^{\mp jk_z z}$$

$$\therefore k_{x1}^2 + \left(\frac{n\pi}{b}\right)^2 + k_z^2 = k_1^2$$

$$k_{x2}^2 + \left(\frac{n\pi}{b}\right)^2 + k_z^2 = k_2^2$$

E_z must be continuous at $x = t$

$$E_z^2 = E_z^1 \text{ at } x = t$$

so, from equation (3),

$$\begin{aligned} \frac{-j\omega\mu_2}{k_2^2} \left(k_2^2 + \frac{\partial^2}{\partial z^2} \right) \psi_2 &= \frac{-j\omega\mu_1}{k_1^2} \left(k_1^2 + \frac{\partial^2}{\partial z^2} \right) \psi_1 \\ \frac{\mu_2}{k_2^2} (k_2^2 - k_z^2) \left[C_n^{2\pm} \sin k_{x2}(a-t) \sin\left(\frac{n\pi}{b}y\right) e^{\mp jk_z z} \right] \\ &= \frac{\mu_1}{k_1^2} (k_1^2 - k_z^2) \left[C_n^{1\pm} \sin(k_{x1}t) \sin\left(\frac{n\pi}{b}y\right) e^{\mp jk_z z} \right] \end{aligned} \quad (7)$$

E_y must be continuous at $x = t$

$$E_y^2 = E_y^1 \text{ at } x = t$$

so, from equation (2)

$$\begin{aligned} \frac{-j\omega\mu_2}{k_2^2} \left\{ C_n^{2\pm} \sin k_{x2}(a-t) \frac{\partial}{\partial y} \left(\sin\frac{n\pi}{b}y \right) (\mp jk_z) e^{\mp jk_z z} \right\} \\ = \frac{-j\omega\mu_1}{k_1^2} \left| C_n^{1\pm} \sin k_{x1}t \frac{\partial}{\partial y} \left(\sin\frac{n\pi}{b}y \right) (\mp jk_z) e^{\mp jk_z z} \right| \end{aligned} \quad (8)$$

Divide (7) by (8)

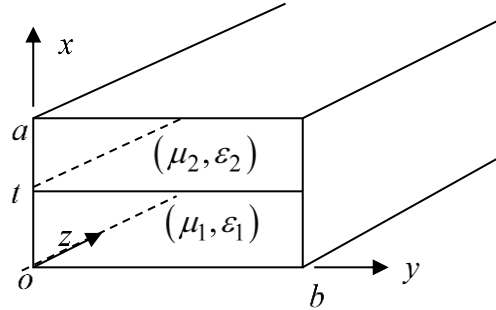
$$\begin{aligned}(k_2^2 - k_z^2) &= (k_1^2 - k_z^2) \\ \therefore k_2^2 &= k_1^2 \quad \text{????}\end{aligned}$$

This indicates that both media have the same k - but this violates the original problem condition $(\epsilon_2 \neq \epsilon_1) \Rightarrow (k_2 \neq k_1)!$

$\therefore$ There are no pure TM_z (or TE_z) modes in this waveguide

Try transverse to x

$$TE_x \Rightarrow \mathbf{F} = \psi \hat{\mathbf{x}}$$



$$\mathbf{E} = -\frac{1}{\epsilon} \nabla \times (\psi \hat{\mathbf{x}})$$

$$E_x = 0 \quad (1)$$

$$E_y = -\frac{1}{\epsilon} \frac{\partial \psi}{\partial z} \quad (2)$$

$$E_z = \frac{1}{\epsilon} \frac{\partial \psi}{\partial y} \quad (3)$$

$$\mathbf{H} = \frac{-j\omega}{k^2} \left(k^2 \psi \hat{\mathbf{x}} + \nabla \frac{\partial \psi}{\partial x} \right)$$

$$H_x = \frac{-j\omega}{k^2} \left(k^2 + \frac{\partial^2}{\partial x^2} \right) \psi \quad (4)$$

$$H_y = \frac{-j\omega}{k^2} \frac{\partial^2 \psi}{\partial y \partial x} \quad (5)$$

$$H_z = \frac{-j\omega}{k^2} \frac{\partial^2 \psi}{\partial z \partial x} \quad (6)$$

$$(\nabla^2 + k^2) \psi = 0$$

$$\psi = X(x)Y(y)Z(z)$$

$$k_x^2 + k_y^2 + k_z^2 = k^2 \quad (7)$$

$$E_z = \frac{1}{\epsilon} X(x) \frac{d}{dy} Y(y) Z(z) \Big|_{y=0}^{y=b} = 0 \quad ; \quad x \in (0, t)$$

$$\Rightarrow Y(y) = \cos\left(\frac{n\pi}{b} y\right)$$

$$\psi_2 = C_2^\pm \sin[k_{x2}(a-x)] \cos\left(\frac{n\pi}{b} y\right) e^{\mp jk_z z}, \quad k_{x2}^2 + \left(\frac{n\pi}{b}\right)^2 + k_z^2 = k_2^2 \quad (a)$$

$$\psi_1 = C_1^\pm \sin(k_{x1}x) \cos\left(\frac{n\pi}{b} y\right) e^{\mp jk_z z}, \quad k_{x1}^2 + \left(\frac{n\pi}{b}\right)^2 + k_z^2 = k_1^2 \quad (b)$$

Enforce B.C. $E_z^2 = E_z^1 \Big|_{x=t}$

from (3)

$$\frac{C_2^\pm}{\varepsilon_2} \sin[k_{x2}(a-t)] \frac{d}{dy} \cos\left(\frac{n\pi}{b} y\right) e^{\mp jk_z z} = \frac{C_1^\pm}{\varepsilon_1} \sin(k_{x1}t) \frac{d}{dy} \cos\left(\frac{n\pi}{b} y\right) e^{\mp jk_z z} \quad (8)$$

Enforce B.C. $H_z^2 = H_z^1 \Big|_{x=t}$

from (6)

$$-C_2^\pm \frac{1}{k_2^2} k_{x2} \cos[k_{x2}(a-t)] (\mp jk_z) e^{\mp jk_z z} = C_1^\pm \frac{1}{k_1^2} k_{x1} \cos(k_{x1}t) (\mp jk_z) e^{\mp jk_z z} \quad (9)$$

Divide (8) by (9)

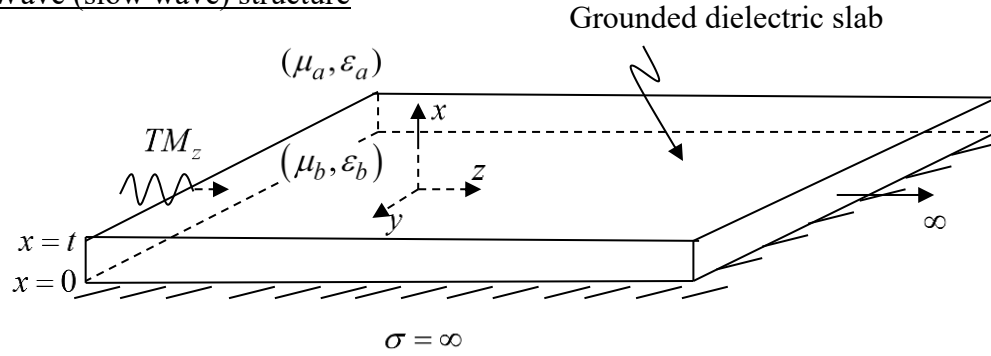
$$\frac{-\mu_2}{k_{x2}} \tan[k_{x2}(a-t)] = \frac{\mu_1}{k_{x1}} \tan(k_{x1}t) \quad (10)$$

$$\text{From (a,b): } k_{x2}^2 - k_z^2 = k_2^2 - k_1^2 \quad (11)$$

(10) and (11) produce a transcendental equation which is used to solve for k_{x1} and k_{x2} , which in turn can be used to solve for k_z , the propagation constant.

$\therefore$ Since there are no contradictions the waveguide mode can be TE_x (also TM_x or a combination of the two).

Surface Wave (slow wave) structure



Consider the TM_z modes in a 2-D grounded dielectric slab, which means the fields have no variation with respect to $y \Rightarrow \frac{\partial Y}{\partial y} = 0$.

B.C.: $E_z(x=0, y, z) = 0$

Radiation condition: Field must decay exponentially to zero as $x \rightarrow \infty$

$\mathbf{A} = \psi \hat{\mathbf{z}}$

$$\mathbf{E} = \frac{-j\omega}{k^2} \left(k^2 \psi \hat{\mathbf{z}} + \nabla \frac{\partial \psi}{\partial z} \right)$$

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \psi \hat{\mathbf{z}}$$

$$E_x = \frac{-j\omega}{k^2} \frac{\partial^2 \psi}{\partial x \partial z}$$

$$H_x = \frac{1}{\mu} \frac{\partial \psi}{\partial y} = 0$$

$$E_y = \frac{-j\omega}{k^2} \frac{\partial^2 \psi}{\partial y \partial z} = 0$$

$$H_y = -\frac{1}{\mu} \frac{\partial \psi}{\partial x}$$

$$E_z = \frac{-j\omega}{k^2} \left(k^2 + \frac{\partial^2}{\partial z^2} \right) \psi$$

$$H_z = 0$$

where $\psi = \psi(x, z) \Rightarrow (\nabla_{xz}^2 + k^2) \psi = 0$

$\psi = X(x)Z(z)$ and $k_x^2 + k_z^2 = k^2$

Identify mode functions:

$\psi_a = C_a e^{-jk_{xa}x} e^{\mp jk_z z} = C_a e^{-j(-j\alpha_a)x} e^{\mp jk_z z} = C_a e^{-\alpha_a x} e^{\mp jk_z z}; x \geq t$ $k_{xa} = -j\alpha_a$ (see pg.11 of notes no.3)

$\psi_b = C_b \sin(k_{xb}x) e^{\mp jk_z z}; 0 \leq x \leq t$

In the region $x \geq t$

$$(a) \quad (-j\alpha_a)^2 + k_z^2 = k_a^2 \Rightarrow -\alpha_a^2 + k_z^2 = \omega^2 \mu_a \epsilon_a$$

In the region $0 \leq x \leq t$

$$(b) \quad k_{xb}^2 + k_z^2 = k_b^2 \Rightarrow k_{xb}^2 + k_z^2 = \omega^2 \mu_b \epsilon_b$$

Enforcing continuity of tangential field components

$$E_z^a = E_z^b \quad \text{at} \quad x = t$$

$$\frac{-j\omega}{k_a^2} C_a (k_a^2 - k_z^2) e^{-\alpha_a t} e^{\mp jk_z z} = \frac{-j\omega}{k_b^2} C_b (k_b^2 - k_z^2) \sin(k_{xb} t) e^{\mp jk_z z}$$

or

$$\frac{1}{k_a^2} C_a (-\alpha_a^2) e^{-\alpha_a t} = \frac{1}{k_b^2} C_b k_{xb}^2 \sin(k_{xb} t) \quad (1)$$

$$H_y^a = H_y^b \quad \text{at} \quad x = t$$

$$-\frac{C_a}{\mu_a} e^{-\alpha_a t} e^{\mp jk_z z} (-\alpha_a) = -\frac{C_b}{\mu_b} \cos(k_{xb} t) e^{\mp jk_z z} k_{xb}$$

$$\frac{C_a}{\mu_a} \alpha_a e^{-\alpha_a t} = -\frac{C_b}{\mu_b} \cos(k_{xb} t) k_{xb} \quad (2)$$

divide (1) by (2)

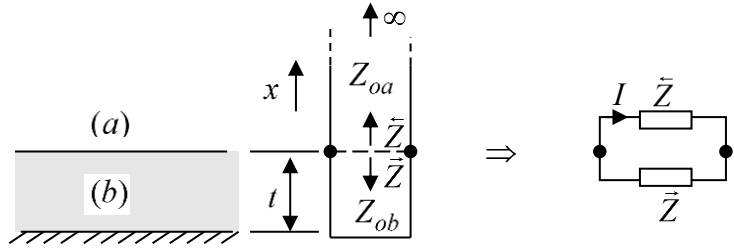
$$\frac{\alpha_a}{\epsilon_a} = \frac{k_{xb}}{\epsilon_b} \tan(k_{xb} t) \quad (3)$$

Substitute (b) into (a)

$$k_{xb}^2 + \alpha_a^2 = k_b^2 - k_a^2 \quad (4)$$

- Transverse Resonance Method

Note: (3) can also be obtained from transmission line techniques.



$$\begin{aligned}
 I\tilde{Z} + I\bar{\tilde{Z}} &= 0 \\
 \tilde{Z} &= -\bar{\tilde{Z}} \\
 Z_{0a} &= -jZ_{0b} \tan(k_{xb}t) \\
 TM \text{ to } x &\Rightarrow \frac{k_{xa}}{\omega\epsilon_a} = \frac{-jk_{xb}}{\omega\epsilon_b} \tan(k_{xb}t) \\
 \frac{\alpha_a}{\epsilon_a} &= \frac{k_{xb}}{\epsilon_b} \tan(k_{xb}t) \quad (3)
 \end{aligned}$$

- Method for solving transcendental equations (3) and (4)

Multiply both sides of (3) by $\epsilon_a t \Rightarrow$

$$\boxed{(\alpha_a t) = \frac{\epsilon_a}{\epsilon_b} (k_{xb} t) \tan(k_{xb} t)} \quad (5)$$

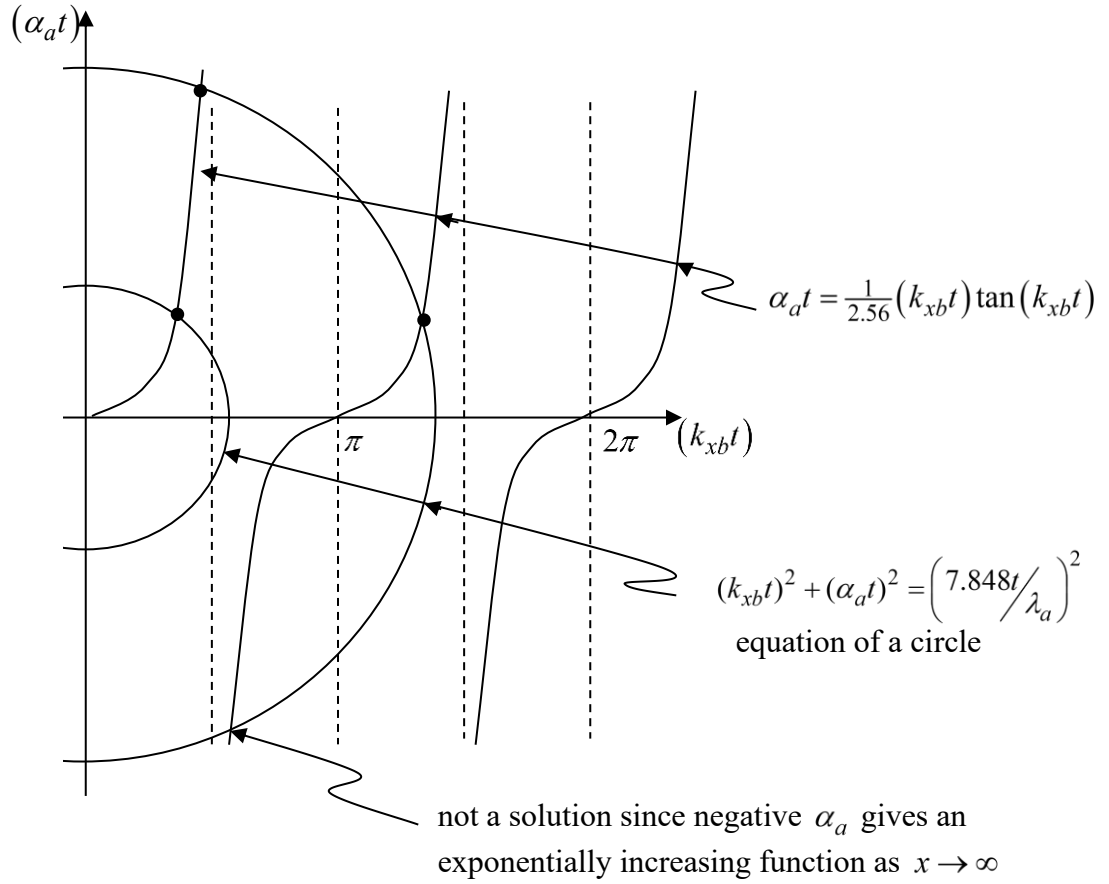
Multiply both sides of (4) by t^2

$$\begin{aligned}
 (k_{xb} t)^2 + (\alpha_a t)^2 &= (k_b t)^2 - (k_a t)^2 = \omega^2 (\mu_b \epsilon_b - \mu_a \epsilon_a) t^2 \\
 &= \omega^2 \mu_a \epsilon_a \left(\frac{\mu_b \epsilon_b}{\mu_a \epsilon_a} - 1 \right) t^2 \\
 \boxed{(k_{xb} t)^2 + (\alpha_a t)^2 &= (k_a t)^2 \left(\frac{\mu_b \epsilon_b}{\mu_a \epsilon_a} - 1 \right)} \quad (6)
 \end{aligned}$$

Since (5) and (6) are two equations with two unknowns, α_a and k_{xb} can be found. This can be done numerically or graphically. The places where these two equations intersect on a graph give α_a and k_{xb} .

Example: let $\mu_b = \mu_a = \mu_o$ and $\varepsilon_b = 2.56\varepsilon_o$, $\varepsilon_a = \varepsilon_o$

Plot (5) and (6) in a Cartesian coordinate system with axes $k_{xb}t$ and $\alpha_a t$.



For $t = \lambda_a/2$ the solns are $\alpha_a t = 3.8$, $k_{xb}t = 1.45$

and at $\alpha_a t = 1.2$, $k_{xb}t = 3.77$

t	α_a	k_{xb}	k_z
$\lambda_a/4$	$6.4/\lambda_a$	$5.04/\lambda_a$	$8.92/\lambda_a$
$\lambda_a/2$	$7.6/\lambda_a$	$2.9/\lambda_a$	$9.86/\lambda_a$
	$2.4/\lambda_a$	$7.54/\lambda_a$	$6.73/\lambda_a$

Note: as t gets smaller, α_a gets smaller so the wave decays more slowly away from the boundary in region a .

Wave velocity in the dielectric slab:

$$k_{xb}^2 + k_z^2 = k_b^2 \Rightarrow k_z^2 = k_b^2 - k_{xb}^2 \leq k_b^2$$

$$-\alpha_a^2 + k_z^2 = k_a^2 \Rightarrow k_z^2 = \alpha_a^2 + k_a^2 \geq k_a^2$$

so

$$k_a^2 \leq k_z^2 \leq k_b^2$$

or

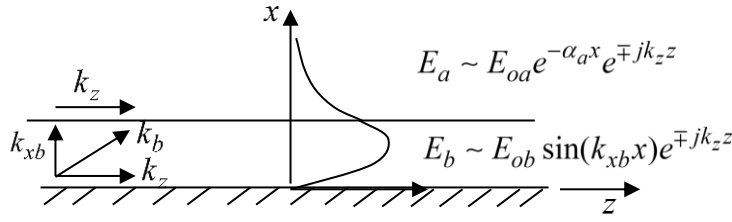
$$\lambda_a \geq \lambda_g \geq \lambda_b \quad ; \lambda_g = \frac{2\pi}{k_z} \text{ guide wavelength}$$

and

$$v_a \geq v_g \geq v_b \quad ; v_g = \frac{\omega}{k_z} \text{ guide phase velocity}$$

The guide phase velocity is less than the velocity of a plane wave in air (infinite homogeneous medium (μ_a, ϵ_a)) so the surface wave is referred to as a slow wave $(v_o > v_g)$.

Note: in region a k_z is real and $k_x = -j\alpha_a$ is purely imaginary so the wave travels in the z direction and decays in the x direction.



(where $k_{za} = k_{zb}$)

In region b , k_x and k_z are both real so the waveguide mode shown in b above breaks up into two plane waves bouncing at the two boundaries as it travels down the waveguide.

Leaky wave

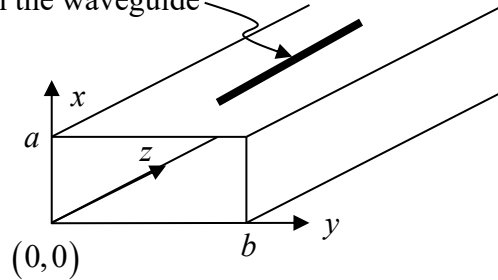
In a closed waveguide

$$k_z = \sqrt{\omega^2 \mu_o \epsilon_o - \left(\frac{n\pi}{a}\right)^2 - \left(\frac{m\pi}{b}\right)^2} \leq k_o \quad (1)$$

so $\lambda_o \leq \lambda_g$, $k_z = \frac{2\pi}{\lambda_g} = \frac{\omega}{v_p}$

and $v_o \leq v_p$ fast wave

If there is a narrow slit in the waveguide then (1) is still a good approximation for k_z in the waveguide



However outside the waveguide k_{zo} and k_{xo} are complex

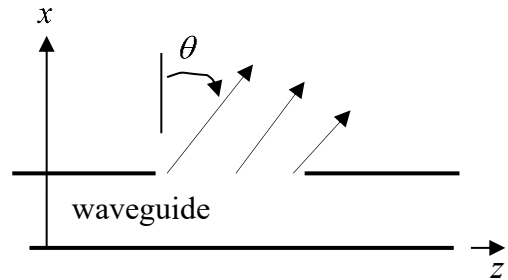
$$k_{zo} = k_z - j\alpha_z \quad (2)$$

same k_z as in the waveguide because of the phase matching requirement

complex part accounts for power leakage outside the guide

Power is “leaked” at an angle θ where

$$\tan \theta \simeq \frac{k_z}{k_x} = \frac{k_z}{\sqrt{k_o^2 - k_z^2 - k_y^2}} \quad (3)$$



If $k_z^2 + k_y^2 = k_o^2 \Rightarrow \theta = 90^\circ$ no power ‘leaked’

$k_z^2 + k_y^2 < k_o^2 \Rightarrow \theta < 90^\circ$ power is ‘leaked’

the leaky wave is said to be a fast wave ($v_g > v_o$)

Ref: Antenna Theory Part 2 Collin and Zucker chapt. 20 by T. Tamir

Appendix: More on Transverse Resonance Method

Find the characteristic equation for a rectangular waveguide.

$$k_x^2 + k_y^2 + k_z^2 - k_o^2 = 0$$

$\uparrow$? $\uparrow$?

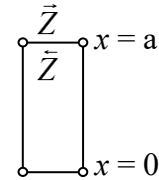
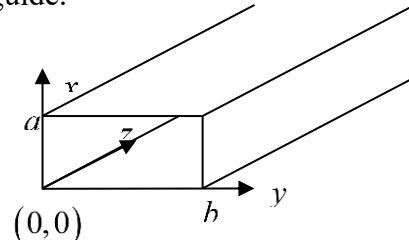
Set

$$\bar{Z} = -\bar{Z}|_{x=a}$$

$$jZ_o^{TE} \tan k_x a = 0$$

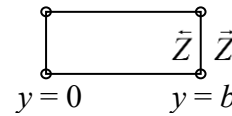
$$\Rightarrow \tan k_x a = 0$$

$$\Rightarrow \sin k_x a = 0 \Rightarrow k_x a = m\pi \text{ or } k_x = \underline{\underline{\frac{m\pi}{a}}}; m=0,1,2,\dots$$



Similarly in the y -direction

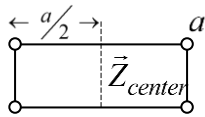
$$\bar{Z} = -\bar{Z}|_{y=b}$$



$$jZ_o^{TE} \tan k_y b = 0 \Rightarrow k_y b = n\pi \text{ or } k_y = \underline{\underline{\frac{n\pi}{b}}}; n=0,1,2,\dots$$

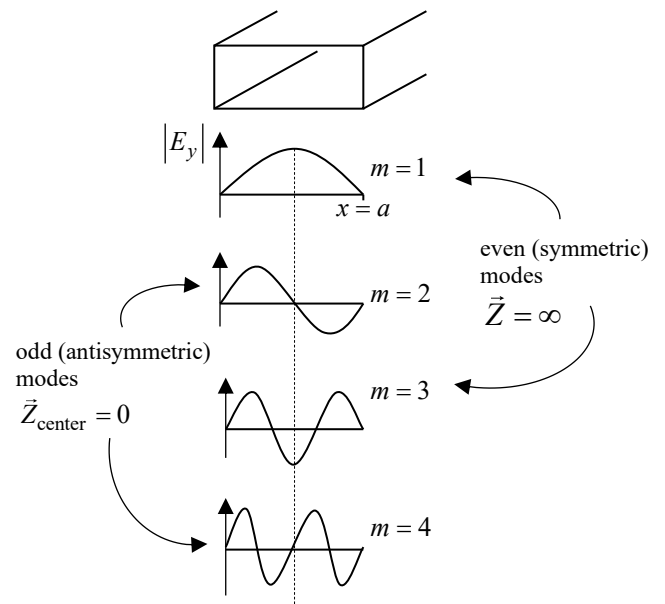
$$\therefore \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + k_z^2 - k_o^2 = 0$$

Look at the impedance at the center of the waveguide



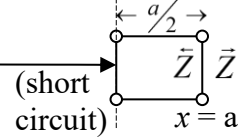
$$\begin{aligned} \bar{Z}_{center} \text{ (or } \bar{Z}_{center}) &= jZ_o^{TE} \tan\left(\frac{m\pi}{a} \frac{a}{2}\right) \\ &= jZ_o^{TE} \tan\left(\frac{m\pi}{2}\right) \\ &= \begin{cases} 0; & m=0,2,4,\dots \\ \infty; & m=1,3,5,\dots \end{cases} \end{aligned}$$

Compare with the electric field distribution



We can use these observations when there is symmetry in the cross-sectional structure of a waveguide.

Example/ return to the rectangular waveguide and find the separation equation by the above alternative approach

odd modes $\Rightarrow \vec{Z}_{center} = 0$ 

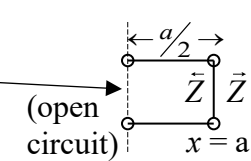
$$\vec{Z} = -\vec{Z}$$

$$jZ_o^{TE} \tan\left(k_x \frac{a}{2}\right) = 0$$

$$\Rightarrow \sin\left(k_x \frac{a}{2}\right) = 0 \Rightarrow k_x \frac{a}{2} = m'\pi$$

$$k_x = (2m') \frac{\pi}{a}; m' = 0, 1, 2, 3, \dots$$

or $k_x = m \frac{\pi}{a}; m = 0, 2, 4, \dots$

even modes $\Rightarrow \vec{Z}_{center} = \infty$ 

$$\vec{Z} = -\vec{Z}$$

$$-jZ_o^{TE} \cot\left(k_x \frac{a}{2}\right) = 0$$

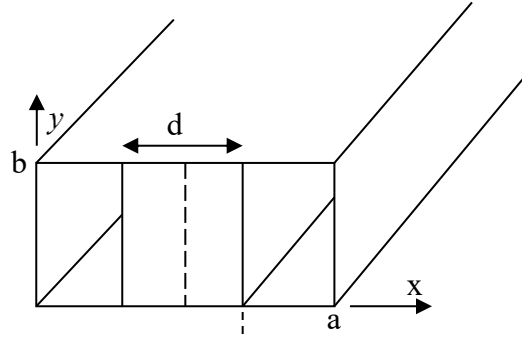
$$\Rightarrow \cos\left(k_x \frac{a}{2}\right) = 0 \Rightarrow k_x \frac{a}{2} = (2m' - 1) \frac{\pi}{2}$$

$$k_x = (2m' - 1) \frac{\pi}{a}; m' = 1, 2, 3, \dots$$

or $k_x = m \frac{\pi}{a}; m = 1, 3, 5, \dots$

$$\therefore k_x(\text{even}) + k_x(\text{odd}) = k_x(\text{total}) = \frac{m\pi}{a}; m = 0, 1, 2, 3, \dots$$

The dominant mode (mode with the lowest cutoff frequency), as far as I know, is always an even node in a rectangular waveguide.



← This is the most convenient place for solving $\vec{Z} = -\vec{Z}$, but any other point in x will give the same solution.

$$\begin{aligned} \vec{Z} &= -\vec{Z} \\ -jZ_{01}^{TE} \cot(k_{x1} d/2) &= -jZ_0^{TE} \tan[k_{x0} (a-d)/2] \\ Z_{01}^{TE} \cot(k_{x1} d/2) &= Z_0^{TE} \tan[k_{x0} (a-d)/2] \\ TE_x \Rightarrow \frac{\omega\mu_1}{k_{x1}} \cot(k_{x1} d/2) &= \frac{\omega\mu_0}{k_{x0}} \tan[k_{x0} (a-d)/2] \\ \text{or } \boxed{\frac{k_{x0}}{\mu_0} \cot[k_{x0} (a-d)/2]} &= \boxed{\frac{k_{x1}}{\mu_1} \tan(k_{x1} d/2)} \\ TM_x \Rightarrow \boxed{\frac{k_{x1}}{\omega\epsilon_1} \cot(k_{x1} d/2)} &= \boxed{\frac{k_{x0}}{\omega\epsilon_0} \tan[k_{x0} (a-d)/2]} \end{aligned}$$

The complete solution would require that the odd mode ($\vec{Z}_{center} = 0$) condition also must be satisfied.